



Homework Questions for Climate and Us Week 2
Due the week of December 1
ANSWER KEY

1. Past Climate

The Antarctic ice sheet, which is about 3 km thick, has yielded ice cores that provide a paleoclimate record extending back to as far as 800,000 years ago. Figure 1 below shows temperature estimates and atmospheric CO₂ data from a 400,000 year-long record from Vostok in East Antarctica. CO₂ data are derived from direct measurement of ancient air bubbles trapped in the ice and temperature data are derived from a proxy: the isotope ratios of frozen water molecules.

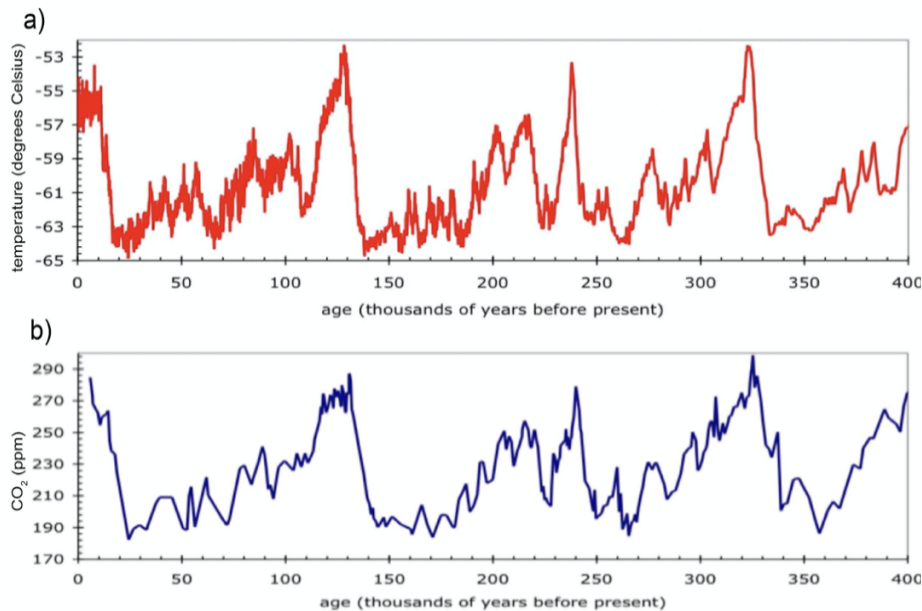


Figure 1. Ice core paleoclimate data for the past 400,000 years from Vostok, Antarctica. a) Variations in temperature (degrees Celsius) vs. time (present day is at 0, age increases to the right). b) Variations in atmospheric CO₂ (ppm) on the same timescale. From Petit J. R. et al. (1999)¹

¹ Petit, J.R., Jouzel, J., Raynaud, D., Barkov, N.I., Barnola, J.M., Basile, I., Bender, M., Chappelaz, J., Davis, M., Delaygue, G., and Delmotte, M. 1999. Climate and atmospheric history of the past 420,000 years from the Vostok ice core, Antarctica. *Nature*, 399(6735), pp. 429-436.

1A. Over the past three million years, Earth's climate has swung between warmer periods and colder "ice ages" during which much of the northern hemisphere was covered in glaciers. From the temperature record in Figure 1a, we can see that there have been four complete ice age cycles in the last 400,000 years, each lasting ~100,000 years on average with somewhat shorter variations superimposed.

1A. (i) What is thought to be responsible for these cycles? What are the time periods of these cycles?

Periodic oscillations of Earth's orbit (so-called Milanković or orbital cycles) influence the distribution of sunlight with latitude. The three most important orbital parameters for climate are eccentricity, obliquity and precession, with periods of about 100,000, 41,000 and 23,000 years, respectively.

1B. Study the Vostok records shown in Figure 1.

1B. (i) How can we most accurately define the relationship between temperature and CO₂ concentration based on Figure 1?

- a. A rise in CO₂ causes a rise in temperature, and a fall in CO₂ causes a fall in temperature
- b. Temperature and CO₂ are positively correlated – they co-vary and rise and fall in cycles
- c. Temperature and CO₂ are negatively correlated – as one rises, the other falls in cycles
- d. A rise in CO₂ causes a decline in temperature, and a fall in CO₂ causes a rise in temperature

B: Temperature and CO₂ have co-varied over the last 400 kyr (thousand years), appearing to rise and fall in cycles. Temperature and CO₂ are positively correlated.

1B. (ii) Based on this relationship, can we conclude that historical changes in climate were driven by changes in atmospheric CO₂ concentration? Why or why not?

No, we cannot. The data depict a correlation, but we cannot definitively infer causation from them.

1B. (iii) What mechanism(s) might explain this relationship?

One mechanism to explain this correlation is the greenhouse gas effect. Greenhouse gases (such as CO₂) trap infrared radiation (heat) in Earth's atmosphere. The incoming energy from the Sun is mainly in the visible spectrum and can penetrate through the atmospheric greenhouse gases. However, these gases are effective at trapping the infrared (thermal) energy output from Earth's surface, resulting in global heating of the Earth.

2. Present Climate

Ice, which covers 10% of Earth's surface, is disappearing rapidly. Climate change has affected glaciers, sea ice, and continental ice sheets worldwide. Let's explore Arctic sea ice in more detail.

2A. Arctic sea ice naturally grows and shrinks throughout the year. However, Arctic sea ice is now declining over time due to warmer temperatures. One way that researchers measure changes in sea ice is through minimum extent, or the area in which at least 15% of the ocean is covered in ice. Figure 2 shows the average monthly Arctic sea ice extent for April from 1980 to 2025.

2A. (i) Which of the following directly contributes to sea level rise? Select all that apply.
Explain your answer in 1-2 sentences.

- a. The melting of land ice sheets in Greenland and Antarctica
- b. The melting of glaciers
- c. The melting of Arctic sea ice
- d. All of the above

A and B. Melting sea ice does not contribute to sea level rise, but melting land ice and melting glaciers will.

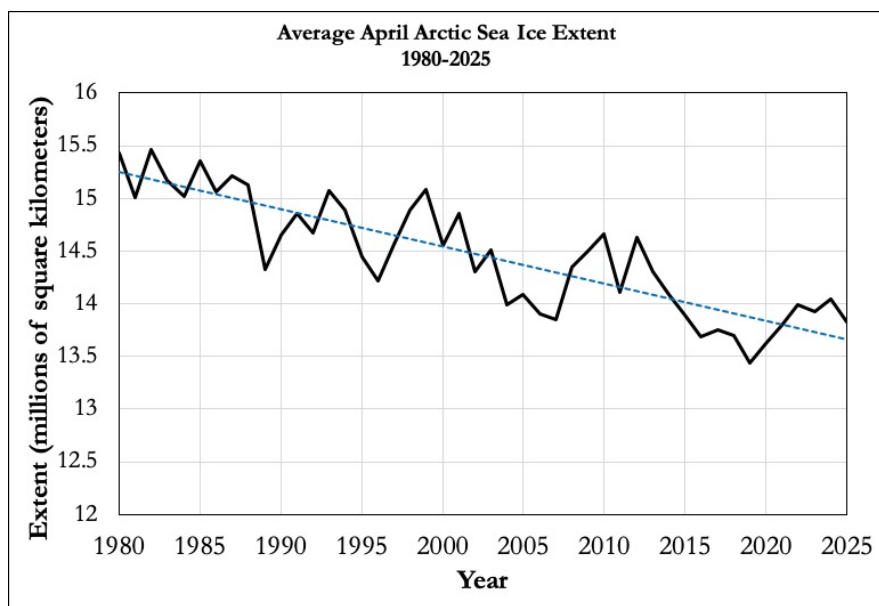


Figure 2. Average monthly April Arctic sea ice extent from 1980 to 2025 in millions of square kilometers. The average is shown in blue. Credit: National Snow and Ice Data Center.

2A. (ii) Using the average line (blue) in Figure 2, **estimate the yearly rate of change of April sea ice over this 46 year period in m^2/year , and choose the answer closest to your estimate.** Show your work! *Note: it is 46 years because we are considering 1980 to 2025, inclusive.*

- a. $-0.0417 \text{ m}^2/\text{year}$
- b. $-41,700 \text{ m}^2/\text{year}$
- c. $-4.17 \times 10^7 \text{ m}^2/\text{year}$
- d. $-4.17 \times 10^{10} \text{ m}^2/\text{year}$
- e. None of the above

$$\frac{13.33 \times 10^6 \text{ km}^2 - 15.25 \times 10^6 \text{ km}^2}{46 \text{ years}} = \frac{-1.92 \times 10^6 \text{ km}^2}{46} = -41,739 \text{ km}^2/\text{year} = -4.17 \times 10^{10} \text{ m}^2/\text{year}$$

2A. (iii) How many square miles of April sea ice has been lost between 1980 and 2025? Choose the answer closest to your calculation AND name an object or place that has approximately this size. Show your work! $1 \text{ km} = 0.621 \text{ miles}$

- a. $6.41 \times 10^4 \text{ miles}^2$
- b. $7.40 \times 10^5 \text{ miles}^2$
- c. $1.45 \times 10^{10} \text{ miles}^2$
- d. $2.33 \times 10^{10} \text{ miles}^2$
- e. None of the above

$$1.92 \times 10^6 \text{ km}^2 \times \left(\frac{0.621 \text{ miles}}{1 \text{ km}} \right)^2 = 7.40 \times 10^5 \text{ miles}^2$$

This is approximately (a little less than) the size of Mexico

2A. (iv) On September 18, 2024 sea ice reached its annual minimum extent of 4.35 million square kilometers, the fourth lowest extent since measurement began in 1978. September sea ice is shrinking at approximately 759,000 km^2 per decade (note that this is September sea ice, not April sea ice that you analyzed in question 2A). If this rate continues, in what year would we expect to have no September sea ice left? Show your work!

- a. 2038
- b. 2081
- c. 2096
- d. 2120
- e. None of the above

759,000 km^2 loss per decade, or 75,900 km^2 per year

$$4.35 \times 10^6 \text{ km}^2 - 75,900 \text{ km}^2 Y = 0, \text{ where } Y = \text{number of years}$$

$$4.35 \times 10^6 \text{ km}^2 = 75,900 \text{ km}^2 Y$$

$$Y = 57.3 \text{ years}$$

$$2024 + 57 = \text{the year } 2,081$$

2B. Melting sea ice requires energy. It takes 5×10^{21} Joules to melt the $16,400 \text{ km}^3$ of sea ice that is lost naturally every year from April to September. Sea ice volume depends on both ice extent and thickness, and is thus an important climate indicator. However, Arctic sea ice volume can not be continuously monitored – instead, numerous measurements and observations from satellites, submarines, and the field are combined into models such as the PIOMAS (Polar Science Center), enabling us to estimate change in sea ice volume over decades

2B. (i) According to PIOMAS, the volume of April sea ice in 2025 was $21.66 \times 10^3 \text{ km}^3$. Using the non-averaged value (black line) from Figure 2, **what was the thickness of April sea ice in 2025 in m?** Show your work! *Hint: Volume = Thickness x Extent*

- a. $1.6 \times 10^4 \text{ m}$
- b. $14 \times 10^6 \text{ m}$
- c. **1.6 m**
- d. $2.3 \times 10^2 \text{ m}$
- e. None of the above

From the figure, the extent of sea ice in 2025 was about $13.8 \times 10^6 \text{ km}^2$

Volume = Thickness x Extent

$$\text{Thickness} = 21.66 \times 10^3 \text{ km}^3 / 14 \times 10^6 \text{ km}^2 = 0.0016 \text{ km} = 1.6 \text{ m}$$

2B. (ii) Assuming the thickness you calculated has not changed, **how much energy would be required to melt the volume of April sea ice that has been lost between 1980 and 2025?** Show your work! *Hint: Utilize your answer in 2Aiii in your calculation, but watch your units*

- a. $3.05 \times 10^{18} \text{ J}$
- b. $3.05 \times 10^{24} \text{ J}$
- c. **$9.37 \times 10^{20} \text{ J}$**
- d. $9.37 \times 10^{22} \text{ J}$
- e. None of the above

$$\text{Volume of ice lost} = .0016 \text{ km} \times 1.92 \times 10^6 \text{ km}^2 = 3,072 \text{ km}^3$$

$$\frac{5 \times 10^{21} \text{ J}}{Y} = \frac{16,400 \text{ km}^3}{3,072 \text{ km}^3}$$

$$Y = 9.37 \times 10^{20} \text{ J}$$

3. Predicting Future Climate

Climate models are important tools that enable scientists to predict future climates and their impacts. Models can be used to predict future CO₂ emissions, temperature, precipitation, sea level rise, etc. In lecture, we focused on models that predicted different possibilities for global fossil fuel emissions that vary based on different emissions scenarios, as well as warming projections. Based on these models, the Intergovernmental Panel on Climate Change (IPCC) in their most recent 2023 report predicted an increase in temperature of 3.2°C (relative to temperature between 1850-1900 CE) by 2100 under current policy measures. Comparing model outputs based on different emission scenarios allows us to assess the potential uncertainty or variability in future climate.

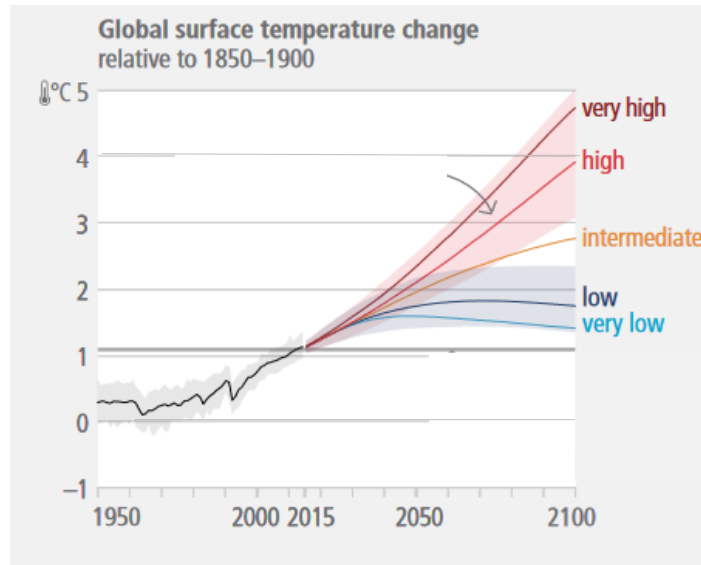


Figure 3. Projected changes in mean global surface temperature under five emissions pathways. The observed mean global temperatures are shown in black, while the model means for the global mean temperature pathways under ‘very low’, ‘low’, ‘intermediate’, ‘high’, and ‘very high’ greenhouse gas emission pathways over the twenty-first century are shown. Uncertainty is shown for the ‘low’ and ‘high’ scenarios. Temperatures are relative to the 1850 – 1900 average.

3A. Figure 3 shows temperature increases predicted by different climate models. **What do the different colored lines represent? What does the shading represent? How does the shading change over time, and what could contribute to this?**

The colored lines represent predicted changes in global surface temperature under various levels of greenhouse gas emissions. The shading demonstrates the uncertainty in these projections for the ‘low’ and ‘high’ emission scenarios. We can see that uncertainty increases as we project further into the future. This is due in part to uncertainty in the importance of different feedback processes.

3B. Based on this figure, what was the global surface temperature in 2015 relative to the 1850-1900 average?

It was 1 degree C warmer than the average.

3C. Based on what you learned about climate models this week, what might lead us to follow the low emission pathway versus the high emission pathway? Name three examples.

Many possible answers. Responses should identify factors that might be expected to reduce emissions over the next 75 years. Examples include decreased reliance on coal or carbon-based energy sources, development of new zero-carbon energy sources, implementation of carbon taxes/pricing, reduced deforestation, changes to building/transportation efficiency standards, etc.

4. Reflections

4A. As you have seen in the different units of Frontiers of Science, converging evidence plays an important role in how scientists provide support for their hypotheses. **Describe how converging evidence relates to climate change using specific examples you learned about in this unit.**

Answers will vary, but students can mention converging evidence of 1) recent increased atmospheric CO₂ in ice cores (this week) and as recorded at Mauna Loa (previous week), 2) past and recent temperature changes in tree ring data with ice core data, or 3) changes in the chemistry of the atmospheric CO₂ (¹³C/¹²C and ¹⁴C/¹²C isotope ratios) coincident with increasing temperatures.

4B. What methods of communication do you think might be most effective when speaking about climate change with someone who considers themselves a denier?

Answers will vary. Respectful and non-accusatory communication is important. By listening and showing curiosity about other people's opinions/preconceived notions, you can engage in a productive dialogue where you can communicate the facts and explain how there is unequivocal evidence of human-caused global warming.

4C. Based on your expected major and career after Columbia, in what ways do you think you might be able to help address climate change and its consequences in the future?

Answers will vary, but for the first part students could say they will decrease their annual carbon footprint by taking fewer flights per year, driving fewer miles per year, taking public transit or biking instead of driving whenever possible, raising the temperature of their air conditioning in the summer by a degree or two, lowering the temperature of their heat in the winter by a degree or two, buying more recycled or previously used products, or a number of other solutions. Answers to the second part will depend entirely on the student's expected major and career.